

On the Implementation and Performance of Swift's Weak References

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CMPSC 263 F25

Links:

- [Video Link](#)
- [Slides Link](#)
- [Writeup Link](#)
- [Github Repo](#)

Build Instructions:

Benchmarks

To run the benchmarks, install [Swift](#) and then run the `runtests_[name].sh` scripts from the benchmarking folder.

SIL Generation

To generate the SIL and LLVM IR code for the tests directory, use the commands:

```
swiftc -emit-sil -O main.swift -o sil.txt
swiftc -emit-irgen -O main.swift -o llvmir_unoptimized.txt
swiftc -emit-ir -O main.swift -o llvmir.txt
```

Refcount Debugging

To run `tests/unsafe_refcount` in debug mode use `swift run --debugger` and then in `lldb` use the command `(lldb) language swift refcount a!` to check the internal reference counts for an object.